

① Standard Normal Distribution

Z-score

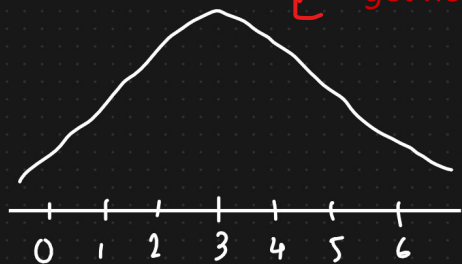
$$X = \{1, 2, 3, 4, 5\}$$

$$\mu = 3$$

Assume 1.41 is 1

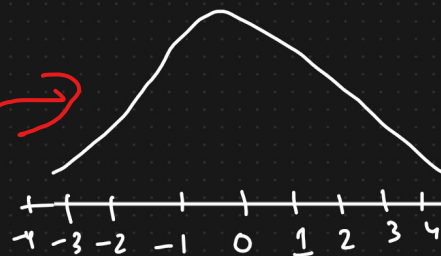
$$\sigma = 1.414 \approx 1$$

We are converting this to that way so that we can get new distribution which mean is 0 and S.D. is 1.



$$\mu = 0$$

$$\sigma = 1$$



$$X = \{1, 2, 3, 4, 5\}$$

$$Z\text{-score} = \frac{x_i - \mu}{\sigma} \quad y: \{-2, -1, 0, 1, 2\}$$

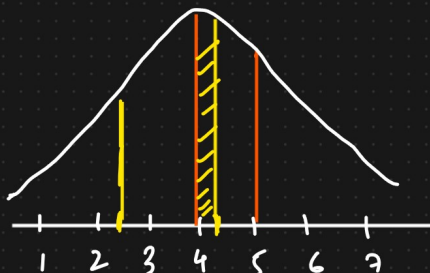
$$\textcircled{1} \quad \frac{1-3}{1} = -2$$

$$\textcircled{3} \quad \frac{3-3}{1} = 0$$

$$\textcircled{2} \quad \frac{2-3}{1} = -1$$

$$\textcircled{4} \quad \frac{4-3}{1} = 1$$

$$X \approx \text{SND}(\mu = 0, \sigma = 1)$$



$$\mu = 4$$

$$\sigma = 1$$

Q.) How many standard deviation 4.25 is away from the mean?

$$x_i = 4.25$$

$$Z\text{-score} = \frac{4.25 - 4}{1} = \underline{\underline{0.25}}$$

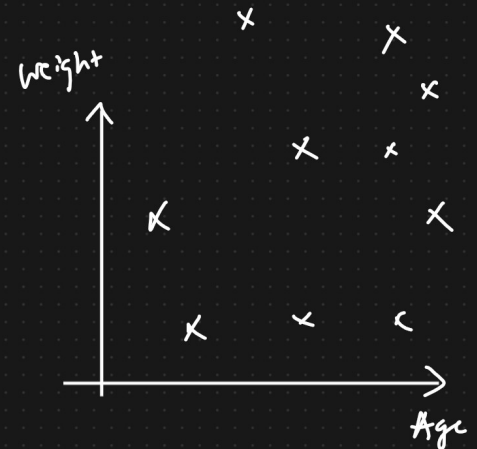
Q.

$$x_i = 2.5$$

$$Z\text{-score} = \frac{2.5 - 4}{1} = \underline{\underline{-1.5}}$$

Eg: Dataset

Age	kg Weight	cm's Height	INR Salary
24	70	175	40K
25	60	160	50K
25	55	180	60K
22	40	130	30K
30	30	175	20K
31	25	180	70K
↓	↓	↓	↓



- ① Clustering Algorithms
- ② Linear Regression
- ③ Logistic Regression

Standardization $\Rightarrow$ ML Models

$$Z\text{-score} = \frac{x_i - \mu_{age}}{\sigma}$$

$$\frac{x_i - \mu_{weight}}{\sigma}$$

In Machine Learning we perform standardization upon all feature and try convert all units same (cm, kg,)

another method is Normalization